

Vizij $\times$ Retico: A System Architecture for Driving Expressive Robot Faces from Incremental Dialogue

Abstract—Effective human–robot interaction is both multimodal and incremental, yet these properties are usually pursued separately: expressive-face systems render gaze and emotion with no real-time signal to drive them, while incremental dialogue systems expose no expressive output. We present a system architecture integrating *retico* [1], an incremental framework for real-time spoken dialogue and robot control, with *Vizij* [2], an open-source framework for designing, animating, and deploying expressive rendered robot faces. Recent systems generate robot facial expression from language models, but do so from completed dialogue state; the architecture described here renders revisable partial signals mid-utterance. Every stage is substitutable, so the architecture serves as a platform for evaluating individual modules rather than as a single demonstration.

Index Terms—expressive rendered faces, incremental dialogue, social robots

I. INTRODUCTION

Effective Human–Robot Interaction (HRI) must move beyond a unidirectional, listen-and-act paradigm toward genuinely bidirectional dialogue. For safe, trustworthy, and engaging interactions, robots must not only interpret instructions but also ask clarifying questions, signal uncertainty, take turns, explain their behavior, and align with human intent. This work contributes directly to that goal by pairing two open-source systems that together span the full loop such interactions require: (1) *retico* [1], an incremental framework for real-time spoken dialogue and robot control, which determines *what* the robot communicates and *when*; and (2) *Vizij* [2], [3], a framework for designing, animating, and deploying expressive rendered robot faces, which determines *how* that communication appears—through gaze, visemes, and emotional expression.

II. MOTIVATION

Effective Human–Robot Interaction is multimodal and incremental. These two properties are usually pursued in silos: expressive-face systems render gaze and emotion agnostic of a signal to drive them, while incremental dialogue systems evaluate robot behavior agnostic of a holistic multimodal robot expression. Interaction that is responsive in time but expressed on a static face—or expressive in the face but lagging a turn behind—fails in the same way, because a human partner reads timing and expression together.

A. Robot Expressivity & Vizij

Dialogue must align with the robot’s appearance and expression. What the robot says is only part of the message, and a face that is static or mistimed undercuts the spoken content. Nonverbal expressions carry social meaning alongside speech,

and a listener integrates both into a single interpretation [4], [5]. When the two diverge, for example a face that stays neutral while the robot apologizes, or a face’s gaze that remains steady while it yields to the dialogue partner, the nonverbal channel does not simply go unused; it contradicts the words and results in a misaligned dialogue. Alignment is therefore not decoration on top of dialogue but a condition for the dialogue being understood as intended.

Expressivity can include a robot face that expresses gaze, visemes, and emotions effectively and in sync. Each channel does distinct work: gaze manages turn-taking, coordinates joint attention, and signals cognitive effort [6]; visemes tie articulation to the speech signal so the face is seen to be speaking [7]; and emotional expression shapes how users judge trustworthiness and remain engaged [8].

Vizij: an open-source ecosystem for designing, animating, and deploying expressive rendered robot faces [2] presents a system that addresses this multichannel communication. It provides a standardized yet modular rigging and controller system spanning end-user definable channels such as eye gaze, visemes, and emotional expression. *Vizij* enables faces authored to be portable across robot platforms rather than rebuilt per deployment. What *Vizij* does not supply is the moment-to-moment signal that should drive those controllers.

B. Effective Human–Robot Spoken Dialogue

Dialogue is incremental. Speakers plan and articulate in overlapping stages rather than composing a whole utterance before speaking [9], and listeners develop their interpretations as words arrive [10]. This is what makes fast feedback, overlap, and repair possible, and it is why turn-taking can be predicted continuously from incremental speech rather than detected after a silence [11], [12]. Systems built following this incremental approach are measurably more responsive and are better received by users [13], [14].

While Large Language Models (LLMs) are increasingly used to generate dialogue, they process data monotonically. LLMs operate over sentence- or multi-sentence-level input and cannot revise an interpretation or a partial output as evidence arrives, which is precisely the capability real-time interaction demands [15]. The consequence for a robot is not merely added latency but a coarser interaction.

Retico: addresses this by formalizing an Incremental Unit (IU) framework in which modules exchange revisable units of information [16]. It presents the IU framework as a modular, real-time pipeline in which modules exchange incremental

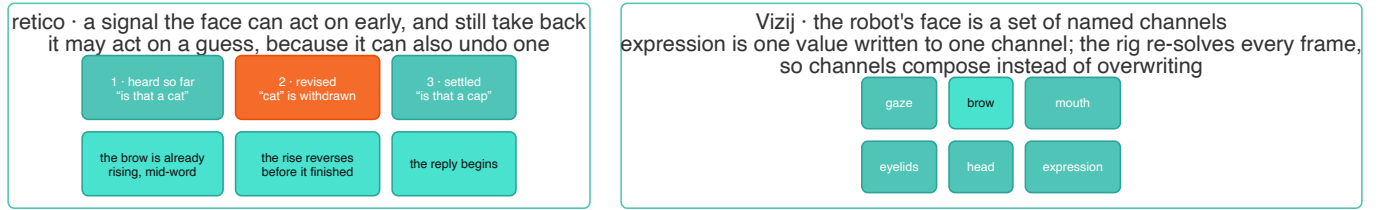


Fig. 1. The two halves. **Left:** retico contributes signals that are revisable — a partial hypothesis can be added, withdrawn, and replaced — so the face may act on a guess because it can also undo one. **Right:** Vizij contributes a face expressed as named channels rather than as a set of pictures; expression is a value written to a channel, and the rig re-solves every frame so channels compose instead of overwriting one another.

units, so the system can act on partial input and revise its decisions as further evidence arrives [1], [16]. Subsequent work extends it toward robots specifically, providing the distributed, time-aligned, multimodal structure that physical deployment requires [17], [18].

C. What already exists

Open-source tooling for rendered faces is not new: FLEX-SDK supports face design, expression authoring, Wizard-of-Oz control and scripting [19], and Furhat has served as a rendered-face platform for over a decade [20]. Nor is model-driven expression: recent systems generate facial expression from language models, per interaction context [21] or during live dialogue [22]. What distinguishes the present work is neither property. Each of those systems consumes *completed* dialogue state—an utterance, a turn, a context—resolved before an expression is chosen. None consumes revisable partial signals, and so none can render a listener response while the partner is still speaking.

III. PROPOSAL: VIZIJ $\times$ RETICO

Existing tooling renders robot faces [19] and generates their expression from language models [21], [22], but each consumes dialogue state that has already resolved, and so cannot respond while the partner is still speaking.

In this paper, we present a system architecture integrating Vizij and Retico, so that the robot's expressive face is driven by the same incremental signals that drive its dialogue. This integration results in three key benefits. The first concerns timing: because expressive signals are available before an utterance concludes, the face can participate during a partner's turn rather than responding after it, which is the behavior that distinguishes conversational presence from sequential exchange. The second concerns revisability: because the provisional status of a signal is preserved rather than discarded at the boundary between systems, the face may act upon a hypothesis precisely because it retains the ability to withdraw one. The third concerns reuse: because the two systems meet through a single specified interface rather than through mutual accommodation, each component within that interface can be replaced without disturbing the others, and the architecture becomes an instrument for comparison rather than a single demonstration.

The proposed interaction pipeline includes the following components:

- *Sensors.* The robot senses its dialogue partner's speech through the microphone and their emotional expression through the camera.
- *Turn-taking.* The turn-taking module continuously estimates when the robot should speak from the audio signal directly [11], [12]. It runs in parallel with recognition rather than downstream of it, so a turn estimate is available before any transcript exists.
- *Incremental speech recognition.* The speech recognition module emits partial hypotheses as it determines what the dialogue partner is saying, labelling each as added, withdrawn, or confirmed, so that downstream modules receive not only the current interpretation but its epistemic standing [1], [16].
- *Incremental dialogue production.* The reply is generated as a stream and split into clauses, each of which is synthesised and sent to the face as it is produced rather than after the reply completes.
- *Emotion generation.* Two sources drive expression. The robot's own affect is declared by the dialogue model at the start of each reply clause, so the face is set as that clause begins to be spoken rather than after the reply ends. Separately, the dialogue partner's expression is estimated from the camera and mirrored faintly.
- *Viseme generation.* Mouth shapes follow the robot's own outgoing speech. Where the synthesiser reports phoneme timings, they drive the mouth poses directly [7], [23]; where it does not, mouth opening follows the amplitude of the playing audio as a fallback.
- *Arbitration and rendering.* Channels that more than one signal wants are assigned to a single occupant by fixed precedence. Gaze is the contested case—turn posture, aversion, and joint attention all want the eyes—while brief impulses such as nods and brow flashes are composed over the sustained posture rather than displacing it [5], [24], [25]. The two emotion sources are ordered rather than mixed: the robot's own declared affect suppresses mirroring of the partner for several seconds, so the face does not reflect the partner's expression while saying something that calls for its own. Visemes and emotional expression need no arbitration, as they drive

separate channels that the rig composes.

We implemented this pipeline in a modular approach that allows the developer / designer / researcher to swap out different modules and evaluate their impact. The significance of this is methodological rather than merely practical. Substituting one component while holding the remainder fixed enables thorough scientific evaluation of the modified component and reproducible comparison with the original. Our proposed architecture is therefore not only a demonstration of a single system but also a platform for future work to evaluate and compare individual modules or module combinations.

A single substitution illustrates the point. Exchanging only the dialogue model, with every other stage fixed, moved median time-to-first-token from 0.40 s to 1.04 s—and the newer model was the slower of the two, because it is served only from a geographically general endpoint whose routing cost exceeded the model’s advantage. Where a model is served can matter more than which model it is, and that is a comparison this architecture makes cheap to run.

We developed this pipeline as a web-based demonstration, using the browser’s microphone and camera in place of the robot’s sensors and rendering the publicly available Quori [26] face.

IV. DISCUSSION

Portability here is a claim about the value stream, not about perception. Display geometry, proportion, and physicality are known to change how a face is read [27], [28], so an identical channel stream rendered on a physical platform is not an identical face. Driving several embodiments from one behaviour pipeline is established practice [29]; establishing that they are *perceived* alike is a separate empirical question.

Three limitations bound what is claimed. The robot’s affect is declared by the model generating its words rather than derived from a model of its own state, and the partner’s expression is estimated categorically and coarsely. Turn-taking and listener-cue quality is inherited entirely from the upstream predictors—this architecture renders those signals, it does not improve them. And there is no user study: every claim here is a claim about the system, not about its effect on people.

V. CONCLUSION

In this paper, we described an architecture that couples incremental spoken dialogue to an expressive robot face. The dialogue system, *retico*, determines what is communicated and when, and the face system, *vizij*, determines how that communication appears. The architecture enables robot expression to begin before an utterance is complete and be withdrawn should the collected evidence change. Both systems we integrate are open source and each module within the pipeline is substitutable, which makes the proposed architecture a testbed rather than a simple demonstration—a form of contribution the field has argued it requires more of [30]. We hope this work will enable future research to evaluate the impact of individual modules or module combinations on the quality of human-robot interaction.

VI. AI USE DISCLOSURE

The authors used generative AI tools to assist in the writing of the paper and identifying relevant references. The authors provided an original outline to limit the AI’s influence on the content then verified and re-wrote the resulting output.

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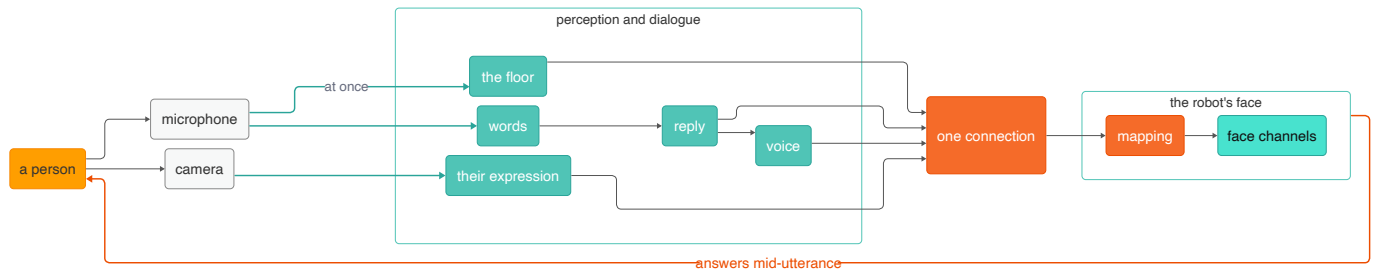


Fig. 2. The pipeline. Microphone audio reaches *the floor* (is a turn ending? is now a moment to nod?) and *words* at the same instant rather than in sequence, which is why a listener response can be rendered before a transcript exists. Every teal stage is a provider chosen at run time — words from an on-board or a cloud recogniser, expression from either of two vision models, reply from any OpenAI-compatible chat model whether local or hosted, voice from any of three synthesisers — and availability is computed from the environment, so an unconfigured provider appears disabled rather than failing when selected. *Mapping* is the editable table that decides which signal moves which channel, and how fast.

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